

Fazel Babaei Rudsari

Unreal Engine
Developer | C++ /
Blueprint | Gameplay
Programming &
Systems



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📁 Professional Experience

PROFESSIONAL SUMMARY

Unreal Engine Developer with 4+ years of experience building gameplay systems in C++ and Blueprints across Mobile, PC, and VR platforms. Experienced in gameplay architecture, debugging, performance optimization, and Unreal Engine-to-backend integration. Strong hands-on experience with Django, REST APIs, authentication, data persistence, leaderboards, and live-service systems. Comfortable taking features from initial prototype through implementation, optimization, testing, and deployment.

PROFESSIONAL EXPERIENCE

Unreal Engine Developer | 09/2021 – 05/2026

- **Unreal Engine:** Developed gameplay systems using C++ and Blueprints from prototyping through deployment.
- **Debugging:** Investigated complex gameplay, Blueprint, C++, build, and engine-level issues.
- **Performance:** Profiled and optimized frame rate, memory usage, draw calls, load times, and streaming behavior across Mobile and PC.
- **Backend integration:** Connected Unreal projects to Django/REST services for authentication, persistence, synchronization, and leaderboard features.
- **Delivery:** Worked across the full development lifecycle and live-service workflows.

📁 Projects

“24” (Mobile Game - Android) | Lead Developer 🔗

- Designed and developed a mobile puzzle game from concept through deployment, including gameplay systems and live-service features.
- **Backend Integration:** Built and integrated a custom Django backend for user authentication, persistent player data, score submission, and leaderboards.
- **Live Operations:** Implemented a server-driven daily challenge system with automated daily challenge generation and player participation tracking.

Car-Garage Tycoon Simulation (Ongoing) | Lead Developer 🔗

- Architecting a complex garage-management simulation using a hybrid **C++ / Blueprint** gameplay architecture.
- **Backend/API:** Integrated Unreal Engine with Django REST services using VaRest for player data synchronization and persistent game-state management.
- Designed modular gameplay systems for garage management, staff progression, orders, property management, and player progression.
- Diagnosed and optimized performance bottlenecks on low-end Android devices, increasing FPS from **28 to 48**.

Strategic PC Game (In Development) | Solo Developer

- Developing a strategy game prototype in Unreal Engine with a focus on systemic gameplay and AI-driven behaviors.
- Building complex UMG interfaces for managing large amounts of gameplay and strategic information.

Technical Projects & Case Studies

First-Person Shooter (FPS) Project | Unreal Engine 5.3

- Collaborated on a team-based FPS title, taking ownership of **AI behavior systems, Animation Blueprints**, and complex gameplay logic.
- Designed and implemented interactive environmental elements, including custom aircraft mechanics and flight control systems using Blueprints.
- Refactored legacy code and optimized performance bottlenecks in the original build, resulting in a significantly smoother and more stable demo.

Third-Person Shooter Development (Solo Project) | UE 5.3

- Developed a complete 3rd-person shooter from scratch, integrating third-party assets with custom-built core systems.
- Architected robust systems including **AnimBlueprints, Character Blueprints**, and **Weapon Systems**, ensuring seamless state transitions and high-fidelity interaction.
- Managed the end-to-end implementation of game logic, from character movement physics to projectile combat systems.

VR Simulation Development | MetaXR & UE 5.6

- Developed immersive VR interactions using MetaXR tools in Unreal Engine 5.6.
- Focused on **user experience optimization**, specifically addressing motion sickness through camera rigging and frame-rate stability techniques suitable for high-end VR hardware.

CORE TECHNICAL SKILLS

- **Unreal** : UE5 · C++ · Blueprints · Gameplay Architecture · Gameplay Systems · UMG
- **Gameplay** : Character Controllers · AI · Combat Systems · Animation Systems · Interaction Systems
- **Backend** : Django · REST APIs · Authentication · Data Persistence · Player Synchronization · Leaderboards
- **Optimization** : CPU/GPU Profiling · FPS Optimization · Memory · Draw Calls · Loading · Streaming · VRAM
- **Integration** : VaRest · Android JNI · Firebase · Android Development
- **Platforms** : Mobile · PC · VR · MetaXR · Meta Quest
- **Tools** : Git · Android Studio · Visual Studio · VS Code · Rider